

Phase 1 (Hindi Version)

Fundamentals — Index & Study Checklist

English:

Goal: build every foundational HLD block from zero -- no prior system design knowledge assumed. Complete chapters in order; each later chapter assumes earlier ones.

Hindi:

लक्ष्य: हर HLD building block को शुरुआत से बनाना -- यह मान लिया गया है कि आपको पहले से कोई system design knowledge नहीं है। Chapters को क्रम में पूरा करें; हर अगला chapter पछिल्लों पर based होता है।

Important Words:

- **Foundational** – बुनियादी

Chapter List (Reference Table – English)

#	Chapter	Time
1	HLD vs LLD What HLD is, how it differs from LLD and coding interviews	1 hr
2	Client-Server Model & Request Lifecycle The request/response model underlying every system	1.5 hr
3	DNS & How the Internet Really Works Domain resolution, TTLs, DNS-based routing	1 hr
4	Networking Essentials TCP vs UDP, HTTP/HTTPS, WebSockets	2 hr
5	APIs & Communication Contracts REST vs gRPC, sync vs async	1.5 hr
6	Databases 101 — Relational ACID, transactions, indexing	2 hr
7	Databases 101 — NoSQL Key-value, document, column-family, graph	2 hr
8	Caching Fundamentals Cache-aside, write-through, write-back, eviction	1.5 hr
9	Load Balancing Basics L4 vs L7, algorithms, health checks	1.5 hr
10	Proxies & CDNs Forward vs reverse proxy, CDN mechanics	1 hr

#	Chapter	Time
11	Object / Blob Storage Basics S3-style storage vs databases vs local disk	1 hr
12	Introduction to Scalability Vertical vs horizontal scaling, statelessness	1.5 hr
13	Non-Functional Requirements & Latency Numbers SLI/SLO/SLA, percentiles, latency intuition	1.5 hr
14	Back-of-the-Envelope Estimation Estimating QPS, storage, bandwidth	1.5 hr
15	Phase 1 Recap & Mini System Capstone: design a pastebin end to end	1.5 hr

Phase 1 total: ~19 hours

Study Checklist

English:

Mark each chapter complete only once you can do all three of the following without notes.

Hindi:

किसी chapter को तभी complete मानें जब आप बिना notes देखे यह तीनों काम कर सकें।

English:

- Explain the core concept out loud in under 2 minutes, using the backend-engineer analogy as a bridge.
- Answer both "Common Interview Questions" from memory with a full model-approach structure.
- Complete the practice exercise(s) unaided.

Hindi:

- Core concept को 2 मिनट से कम में ज़ोर से बोलकर समझाएं, backend-engineer analogy को पुल की तरह use करते हुए।
- दोनों "Common Interview Questions" का जवाब याद से, पूरे model-approach structure के साथ दें।
- Practice exercise(s) बिना किसी सहायता के पूरे करें।

Important Words:

- **Bridge (figurative)** – पुल / जोड़ने वाली कड़ी

How Phase 1 Chapters Connect

English:

- Ch 1 sets the frame (HLD vs LLD) so every later chapter is understood at the right altitude.
- Ch 2-5 build the communication layer: how a request travels, over what protocols, in what contract shape.
- Ch 6-7 build the storage layer: where data lives and which guarantees it has.
- Ch 8-11 build the performance/delivery layer: caching, load balancing, proxies/CDN, object storage.
- Ch 12-14 build the scale-reasoning layer: how to grow the system and how to quantify 'how much.'
- Ch 15 forces you to combine all of the above into one working design.

Hindi:

- Ch 1 पूरा frame सेट करता है (HLD vs LLD) ताकि बाकी हर chapter सही altitude पर समझा जा सके।
- Ch 2-5 communication layer बनाते हैं: request कैसे travel करती है, कसि protocol पर, कसि contract shape में।
- Ch 6-7 storage layer बनाते हैं: data कहाँ रहता है और उसकी guarantees क्या हैं।
- Ch 8-11 performance/delivery layer बनाते हैं: caching, load balancing, proxies/CDN, object storage।
- Ch 12-14 scale-reasoning layer बनाते हैं: system को कैसे बढ़ाएं और 'कतिना' को कैसे measure करें।
- Ch 15 आपको यह सब मिलाकर एक काम करने वाला design बनाने पर मजबूर करता है।

Important Words:

- **Altitude (figurative)** – स्तर / ऊंचाई
- **Quantify** – मात्रा में मापना

Next step

English:

Once every chapter is checked off and Chapter 15's capstone is solved unaided, you're ready to approve Phase 2 (Intermediate).

Hindi:

जब हर chapter check हो जाए और Chapter 15 का capstone बिना सहायता के हल हो जाए, तब आप Phase 2 (Intermediate) approve करने के लिए तैयार हैं।

Important Words:

- **Capstone** – कैपस्टोन / अंतिम बड़ा प्रोजेक्ट

General Words Glossary

A consolidated list of the important English words used throughout this chapter, with Hindi meanings and simple explanations — useful for revising vocabulary after finishing the chapter.

English Word	Hindi Meaning	Simple Explanation
Altitude (figurative)	स्तर / ऊंचाई	The level of detail you're reasoning at
Bridge (figurative)	पुल / जोड़ने वाली कड़ी	Something that connects two ideas
Capstone	कैपस्टोन / अंतमि बड़ा प्रोजेक्ट	A final project that combines everything learned
Foundational	बुनियादी	Forming the base or starting point
Quantify	मात्रा में मापना	To express as a number